

Week 3: Hash tables

A hash table maps keys to values through a hash function.

Average lookup, insert and delete take constant time.

Collisions

Two keys collide when they hash to the same bucket.

Chaining keeps a list per bucket. Open addressing probes for the next free slot.

Load factor

Load factor is entries divided by buckets.

Example: 6 entries in 8 buckets gives a load factor of 0.75, so the table doubles to 16 buckets.